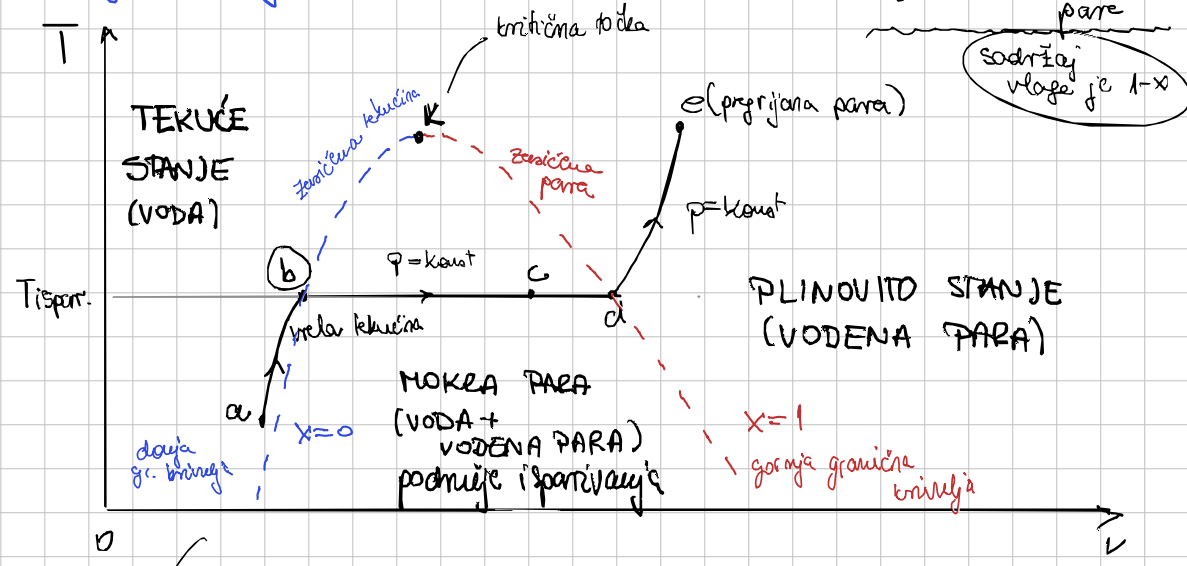
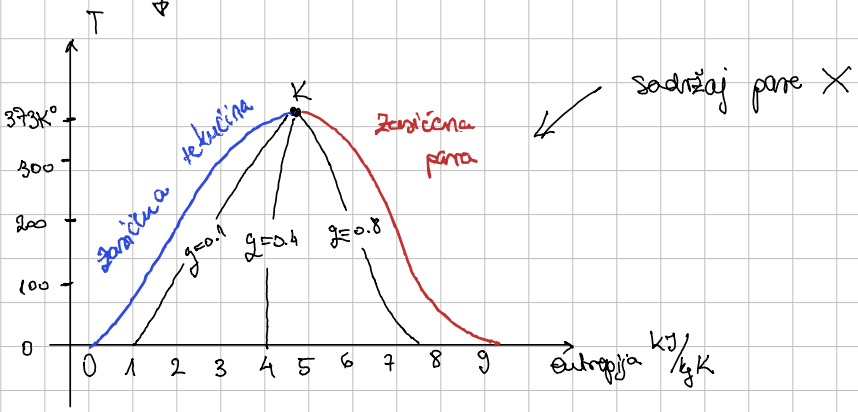
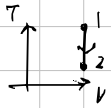
Rankineov kružni proces

Krivulja zasićenja vode





* izentropa



Termoelektrana s parnom turbinom

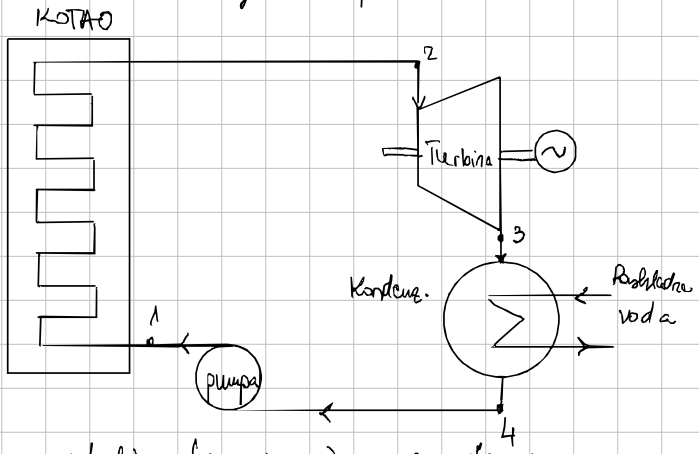
- Rankinov KP → 2 izobarne + 2 adijabatske promjene (u idealnom slučaju izentropske)

- Kotao: p=konst, T↑

  turbina: v↑ (adijabat)

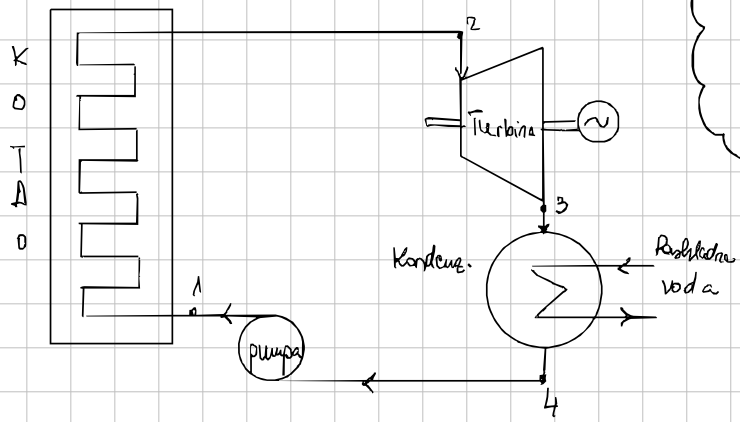
  kondenzator: p=konst., T↓

  pumpa: v↓ (adijabat)



- unutrašnji stupnjevi djelovanja pumpe i turbine (η_p i η_t) → realni ili idealni procesi
- idealan proces u pumpi/turbini → izentropski proces → povratljiv pro

T-s i h-s dijagram

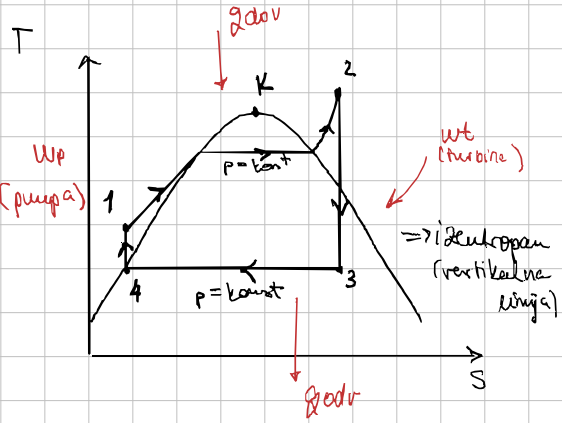


T-s => temp - specifična entropija

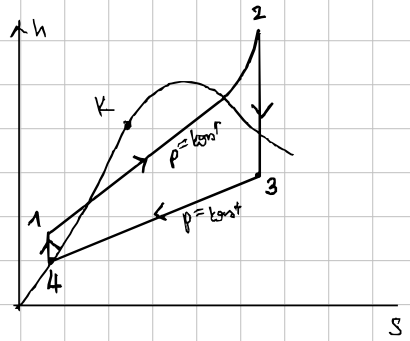
- promjena temp i entropije radnog fluida kroz termodin. proces

h-s Mollierov dijagram entalpija - specifična entropija

- dovođenje topline u kotlu (+)
- dobivanje rada u turbini (+)
- dovođenje topline u kond (-)
- ulaganje rada u pumpi (-)



bolja za razumijevanje TOPLINSKIH procesa

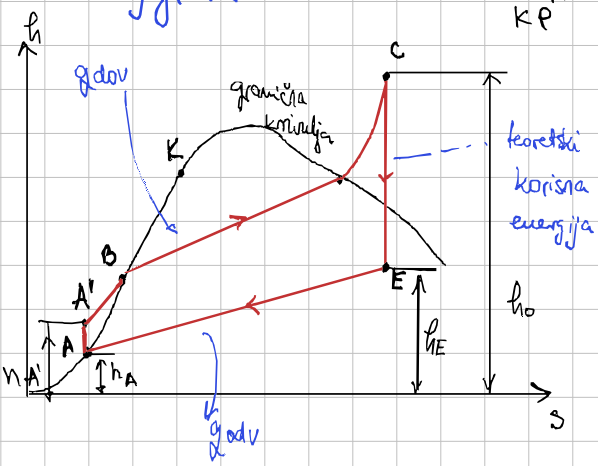


- koristi se puno kod proračuna i analize turbina i kompresora
  ↳ moguće direktno očitati energije i učinkovitost

praktičniji za račun i promjenu energije / mehaničkog rada strojeva

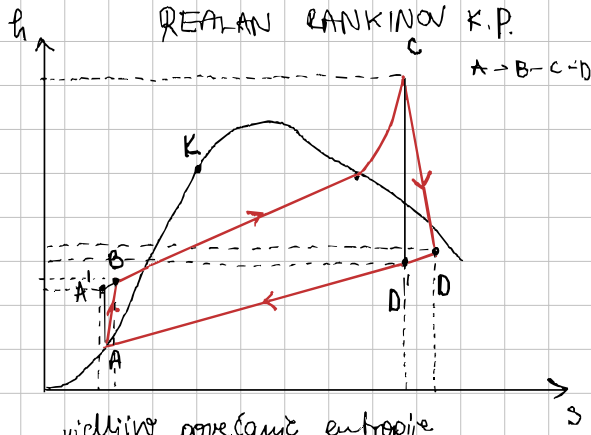
h-s dijagram

IDEALAN RANKIN KP



REALAN RANKINOV K.P.

A → B – C – D

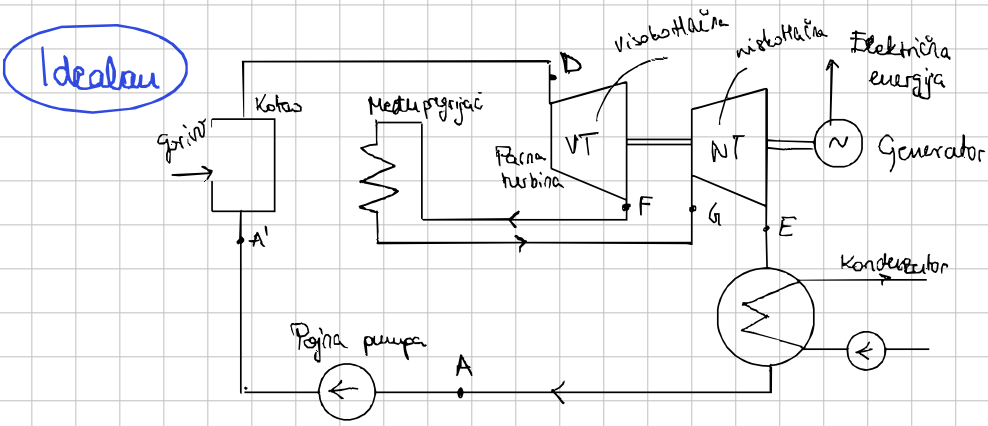


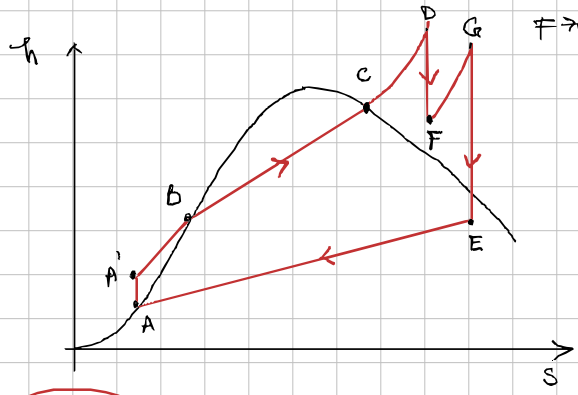
vidljivo povećanje entropije (A – B i C – D)

=> nepovratljivost procesa

RANKINOV PROCES S MEĐUPREGRIJANJEM

Idealan



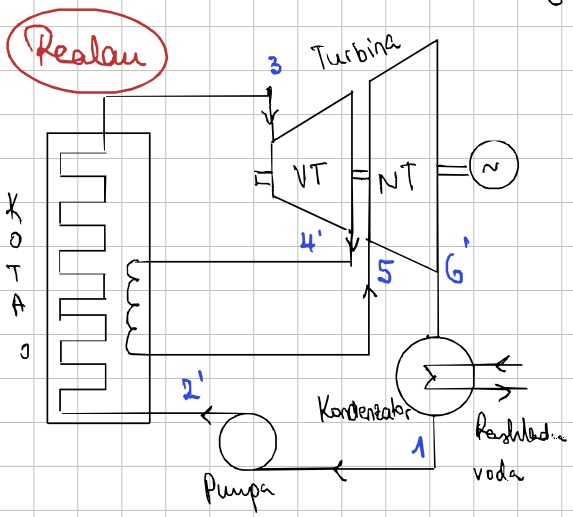


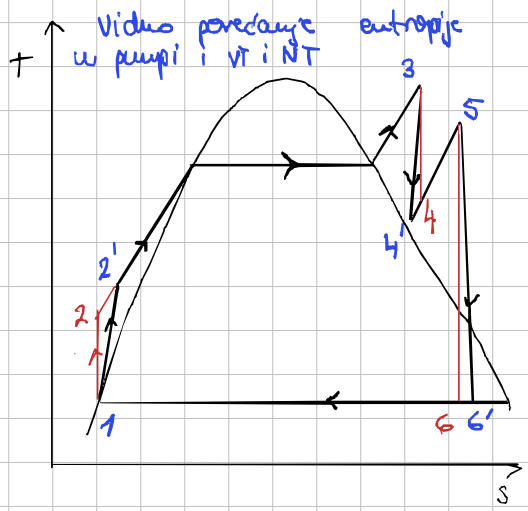
F→G = međupregrijanje

idealan = izentropski u D→F i G→E i A→A' (turbina i pumpa)

• proces dovođenja topline u međupregrijaču je također izobaran

Realan





Termički stupanj djelovanja

• paziti na to je li proces kompresije / ekspanzije idealan ili realan (jesu li zadane umnožaje efikasnosti pumpe i turbine), radi li se o procesu s međupregrijanjem ...

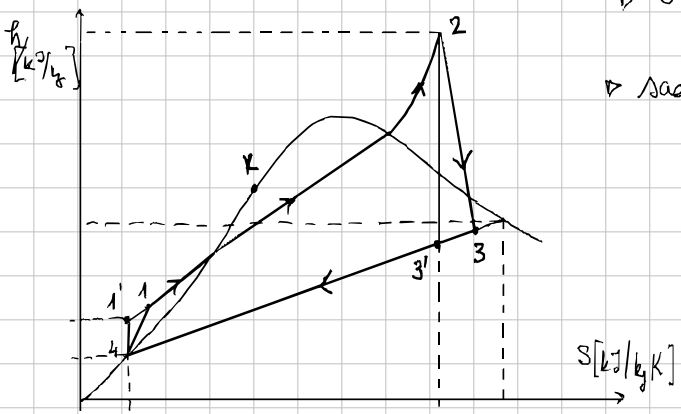
RAD PUMPE:

uzimamo da je tekuća voda nekompresibilna → konstantna gustoća ρ →

→ veza sa specifičnim volumenom → tehnički rad pumpe ($v \Delta p$)

$$\eta_t = \frac{w}{q_{dov}} = \frac{w_t - |w_p|}{q_{dov}}$$

Računanje sadržaja pare x



▷ s' => lijeva granična , s'' => desna granična krivulja

▷ sadržaj pare na izlazu iz turbine x_3 i x_3'

$$h = h' + x(h'' - h')$$

$$u = u' + x(u'' - u')$$

$$v = v' + x(v'' - v')$$

$$s = s' + x(s'' - s')$$